

Mutual Fund Order Management System

Functional Specification for Wealth Management Division

Version 1.1 | Classification: Internal / Engineering Use Only

Scope	USD-denominated mutual fund orders	Fund Types	Open-ended (EOD NAV) and Closed-ended (real-time)
Transfer Agents	NSCC (onshore, 4:00 PM ET) and RBC (offshore, 1:00 AM ET)	Audience	Engineering Team

1. Introduction and Scope

The MF-OMS is a centralized platform for end-to-end processing of mutual fund orders within the Wealth Management division. All orders are denominated in USD and originate from clients, financial advisors, and portfolio management systems.

The system supports two fund categories: (1) **Open-ended funds**, priced at end-of-day NAV and processed via batch workflows; and (2) **Closed-ended funds**, exchange-priced and processed in real time. The primary objective is accurate order execution, aggregation-driven cost reduction, and full regulatory compliance.

2. Order Ingestion

2.1 Sources

- Client digital platforms (web and mobile)
- Financial advisor portals (web and mobile)
- Portfolio Management Systems (PMS) via automated electronic feeds

Both real-time API and scheduled batch ingestion are supported.

2.2 Mandatory Order Attributes

Every inbound order must carry: client account ID, fund identifier (CUSIP/ISIN), order type (Buy/Sell), order amount (USD), trade date, and originating channel. A unique immutable Order ID is assigned upon receipt.

3. Order Validation and Enrichment

3.1 Validation Rules

Validation Check	Failure Action
Mandatory field completeness	Reject with error code; log for ops review
Client account eligibility	Reject; notify originating channel
Fund eligibility and active status	Reject; notify originating channel
Trade date and cutoff compliance	Reject or queue for next cycle
Duplicate order detection	Flag and log; block further processing

3.2 Order Enrichment

- Fund family assignment from fund master reference data
- Transfer agent identification (NSCC for onshore, RBC for offshore)
- Settlement attributes (T+1 per SEC Rule 15c6-1)
- Onshore or offshore fund classification
- Fund cutoff time lookup for transmission scheduling

Enriched orders are persisted in the OMS with status **VALIDATED**.

4. Order Aggregation and Transfer Agent Routing

4.1 Aggregation Logic

Individual client orders are aggregated into Bulk Orders by grouping on: **mutual fund identifier + transfer agent + trade date**. The bulk order dollar value equals the sum of all constituent client order amounts, reducing external transmissions and lowering transfer agent processing fees.

4.2 Transfer Agent Routing

Fund Classification	Transfer Agent	Cutoff (ET)
Onshore Mutual Funds	NSCC (Natl Securities Clearing Corp)	04:00 PM
Offshore Mutual Funds	RBC (Royal Bank of Canada)	01:00 AM

Routing configuration is externalized as reference data to support future transfer agent additions without code changes.

4.3 Transmission Requirements

- Each message includes: aggregated amount, fund ID, trade date, firm ID
- Exactly-once delivery via idempotency keys on all outbound calls
- Configurable retry with exponential back-off for network failures
- Pre-transmission duplicate detection is mandatory
- Transmission timestamp, acknowledgement, and status logged for audit

4.4 Confirmation Processing

Transfer agents return confirmations acknowledging bulk order receipt. The OMS reconciles each confirmation against the transmitted order, verifying amount, fund ID, and trade date. Mismatches are flagged as **reconciliation breaks** and routed to operations within the 1-hour SLA.

5. Contract Processing and NAV Allocation

5.1 Contract Data

After processing all participating firms, transfer agents determine the fund NAV and transmit contract messages back to the Wealth Management division. Each contract includes:

- Net Asset Value (NAV) per share
- Total shares allocated for the aggregated order
- Contract reference identifier

5.2 Share Allocation

The OMS allocates shares to each constituent client order proportional to its dollar contribution to the bulk order total. Each individual order is updated with the applicable NAV, allocated share count, and contract reference.

5.3 Contract Reconciliation

Contract reconciliation occurs between the transfer agents and OMS to ensure NAV values and share allocations match transmitted aggregated orders. Discrepancies are logged as exceptions for operations resolution.

6. Trade Booking and Position Updates

Post-allocation, the OMS generates individual trade records and transmits them to downstream booking systems, which update client portfolios and produce settlement instructions and trade confirmations.

Status Notifications

- Execution and completion notifications pushed to clients, advisors, and PMS
- Real-time updates for closed-ended fund orders
- EOD batch notifications for open-ended fund orders

7. System Constraints

Constraint	Requirement
Exactly-Once Processing	Duplicate detection at every stage (ingestion, aggregation, transmission, contract). Idempotency keys required on all external API calls to transfer agents and booking systems.
Order Immutability	Orders transmitted to transfer agents are write-locked. Only explicit cancellation workflows may alter state; direct modification of transmitted orders is prohibited.
Dual-Mode Processing	Batch aggregation for open-ended funds (EOD NAV workflow) and real-time streaming for closed-ended funds (exchange-priced) must coexist in the same platform.
Transmission Windows	NSCC cutoff is 04:00 PM ET. Orders not aggregated by this time are deferred to the next business day. RBC cutoff is 01:00 AM ET; offshore orders must be validated and aggregated before this window.

Reconciliation SLA	All bulk orders must match transfer agent confirmations within 1 hour. Unmatched orders trigger automated exception workflows and are escalated to the operations team.
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8. Non-Functional Requirements

Attribute	Requirement
Availability	99.9% uptime, 9:30 AM to 4:00 PM ET; active-active failover
Performance	Order ingestion p99 under 500 ms; bulk transmission under 2 s
Security	TLS 1.2+ in transit; AES-256 at rest; RBAC enforced; no PII stored beyond operational necessity
Audit	Immutable audit log for all state transitions; 7-year retention per FINRA Rule 4511
Scalability	Horizontal scaling for PMS batch spikes; queue-based ingestion
Observability	Real-time dashboard: volumes, latency, recon breaks, TA SLAs

9. Exception Management

The OMS provides operations tooling for the following exception types. All events are logged with full order context, resolution steps, and operator identifiers to maintain a complete audit trail.

Exception Type	Handling
Duplicate order at ingestion	Flag and log; block processing
Reconciliation mismatch	Break raised; ops team notified
NAV discrepancy	Contract held; ops investigation required
Missing confirmation (1-hr SLA)	Auto-escalate to exception queue
Transmission failure (retries exhausted)	Alert raised; manual retry workflow

10. Glossary

Term	Definition
NAV	Net Asset Value, the price per unit/share of a mutual fund calculated at end of trading day
NSCC	National Securities Clearing Corporation, transfer agent for onshore mutual funds
RBC	Royal Bank of Canada, transfer agent for offshore mutual funds
Bulk Order	Aggregated client orders for a single fund transmitted as one message to a transfer agent
Contract	Message from transfer agent containing NAV and share allocation for a bulk order
Transfer Agent	Entity receiving fund orders and returning NAV and share allocations
PMS	Portfolio Management System, the automated upstream order source
OMS	Order Management System, i.e., this system (MF-OMS)
Reconciliation Break	Discrepancy between a transmitted order and the transfer agent confirmation